

DSC-520 Final Project Step 3

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2024-11-17

Titanic Survival Analysis: Investigating Socioeconomic Factors on Survival

Introduction

The RMS Titanic stands out as one of history's most notorious oceanic disasters. Occurring on April 15, 1912, the story of the Titanic provides a glimpse into the socioeconomic conditions of the early 20th century, and how various factors may have played a role in the ship's 2,224 passengers' outcomes. Of that number, only about 700 passengers survived.

Data has been collected from the Titanic disaster, which may reveal socioeconomic patterns and shed light onto how social class, gender, and economic status may have influenced the passengers' survival outcomes.

Problem Statement

To understand the role, if any, of socioeconomic factors on the survival outcomes of the passengers aboard the Titanic.

Approach

Our study will follow these steps:

1. Determine a list of data-driven questions, whose answers would shed light on the problem statement.
2. Identify the most reliable data source for performing the analysis.
3. Attain and clean the data, performing the necessary preprocessing for this study.
4. Perform exploratory data analysis to answer the questions identified earlier.
5. Recommend a statistical model which would be effective at addressing the problem statement.

This approach aims to quantify the impact of the known socioeconomic variables of Titanic passengers on their survival outcomes. Our recommended model will aim to identify specific groups and patterns in the data which contributed to the survival outcomes.

Research Questions

1. What was the economic distribution of passengers aboard the Titanic?
2. Does passenger class correlate with survival?
3. Does ticket fare correlate with survival?
4. Does family size correlate with survival?
5. Did a particular age group have the highest survival rate?

Recommended Model

I recommend building a logistic regression model to address the problem statement. Logistic regression is well suited for this task because it is a binary classification model and our problem statement aims to understand the factors at play which could have influenced survival (a binary outcome).

Furthermore, logistic regression can be used with multiple predictors and help us understand the relative impact of each predictor on the outcome. For this, I would suggest the following predictors:

- Passenger class
- Sex
- Fare
- Family size
- Age

Including variables such as sex and age, which are not directly related to socioeconomic status, will give us context for assessing the relative impact of socioeconomic variables, such as passenger class and fare.

Analysis

After some research, I found the following data sources which could be useful for this study:

1. The Titanic Dataset provided by Stanford:
 - This dataset is provided by Kaggle, and was slightly cleaned by Stanford for some course materials
 - Link: <https://web.stanford.edu/class/archive/cs/cs109/cs109.1166/problem12.html>
 - 887 passengers
 - Variables: Survival indicator, passenger class, name, sex, age, number of siblings aboard, number of parents or children aboard, fare paid
2. The dataset extended by data from Wikipedia:
 - Source: <https://www.kaggle.com/datasets/pavlofesenko/titanic-extended>
 - Additional variables: hometown, location boarded, destination, and lifeboat.
3. Titanic Crew Records (British National Archives):
 - Source: <https://discovery.nationalarchives.gov.uk/details/r/C12378516>
 - Information about crew members' roles and outcomes

Since Wikipedia's information is crowdsourced, it is not as reliable as Stanford or the British National Archives. After checking the British National Archives, I found that the data is comprehensive, but would need significantly more preprocessing and wrangling to convert it into a usable format. The Stanford data has already been curated and mostly cleaned for statistical analysis. Also, several other socioeconomic studies have utilized it before. Therefore, I've deemed the Stanford dataset as the most reliable for this study.

Data cleanup

First, let's load the required packages as well as our dataset:

```
# Load required packages
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.1      v tibble   3.2.1
## v lubridate  1.9.3      v tidyr    1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(dplyr)
library(stats)
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(survival)
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:survival':
##
##   cluster
##
## The following object is masked from 'package:purrr':
##
##   lift
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
# Read the data
titanic_raw <- read.csv("titanic.csv", stringsAsFactors = FALSE)
```

Then, we check out the dataset's structure and look out for potential issues which may need cleaning:

```
str_report <- str(titanic_raw)
```

```
## 'data.frame': 887 obs. of 8 variables:
## $ Survived : int 0 1 1 1 0 0 0 0 1 1 ...
## $ Pclass : int 3 1 3 1 3 3 1 3 3 2 ...
## $ Name : chr "Mr. Owen Harris Braund" "Mrs. John Bradley (Florence Briggs Thayer
## $ Sex : chr "male" "female" "female" "female" ...
## $ Age : num 22 38 26 35 35 27 54 2 27 14 ...
## $ Siblings.Spouses.Aboard: int 1 1 0 1 0 0 0 3 0 1 ...
## $ Parents.Children.Aboard: int 0 0 0 0 0 0 0 1 2 0 ...
## $ Fare : num 7.25 71.28 7.92 53.1 8.05 ...
```

```
missing_values <- colSums(is.na(titanic_raw))
summary_initial <- summary(titanic_raw)
```

```
print("Initial Data Summary:")
```

```
## [1] "Initial Data Summary:"
```

```
print(paste("Number of rows:", nrow(titanic_raw)))
```

```
## [1] "Number of rows: 887"
```

```
print(paste("Number of columns:", ncol(titanic_raw)))
```

```
## [1] "Number of columns: 8"
```

```
print("Missing values by column:")
```

```
## [1] "Missing values by column:"
```

```
print(colSums(is.na(titanic_raw)))
```

```
##           Survived           Pclass           Name
##           0           0           0
##           Sex           Age Siblings.Spouses.Aboard
##           0           0           0
## Parents.Children.Aboard           Fare
##           0           0
```

No missing values have been found in the dataset. However, it is still necessary to convert categorical variables into factors:

```
# Convert categorical variables to factors and store the transformed data as a new variable
titanic_clean <- titanic_raw %>%
  mutate(
    Survived = factor(Survived, levels = c(0, 1), labels = c("No", "Yes")),
    Pclass = factor(Pclass, levels = c(1, 2, 3), labels = c("First", "Second", "Third")),
```

```

    Sex = factor(Sex)
  )

# Display the transformed dataset
str(titanic_clean)

## 'data.frame':    887 obs. of  8 variables:
##  $ Survived      : Factor w/ 2 levels "No","Yes": 1 2 2 2 1 1 1 1 2 2 ...
##  $ Pclass        : Factor w/ 3 levels "First","Second",...: 3 1 3 1 3 3 1 3 3 2 ...
##  $ Name          : chr  "Mr. Owen Harris Braund" "Mrs. John Bradley (Florence Briggs Thayer"
##  $ Sex           : Factor w/ 2 levels "female","male": 2 1 1 1 2 2 2 2 1 1 ...
##  $ Age           : num  22 38 26 35 35 27 54 2 27 14 ...
##  $ Siblings.Spouses.Aboard: int  1 1 0 1 0 0 0 3 0 1 ...
##  $ Parents.Children.Aboard: int  0 0 0 0 0 0 0 1 2 0 ...
##  $ Fare           : num  7.25 71.28 7.92 53.1 8.05 ...

head(titanic_clean)

```

```

##   Survived Pclass                                Name    Sex Age
## 1      No   Third                                Mr. Owen Harris Braund   male 22
## 2     Yes   First Mrs. John Bradley (Florence Briggs Thayer) Cumings female 38
## 3     Yes   Third                                Miss. Laina Heikkinen female 26
## 4     Yes   First      Mrs. Jacques Heath (Lily May Peel) Futrelle female 35
## 5      No   Third                                Mr. William Henry Allen   male 35
## 6      No   Third                                Mr. James Moran     male 27
##  Siblings.Spouses.Aboard Parents.Children.Aboard    Fare
## 1                                1                    0 7.2500
## 2                                1                    0 71.2833
## 3                                0                    0 7.9250
## 4                                1                    0 53.1000
## 5                                0                    0 8.0500
## 6                                0                    0 8.4583

```

1. Exploring the economic distribution of Titanic passengers

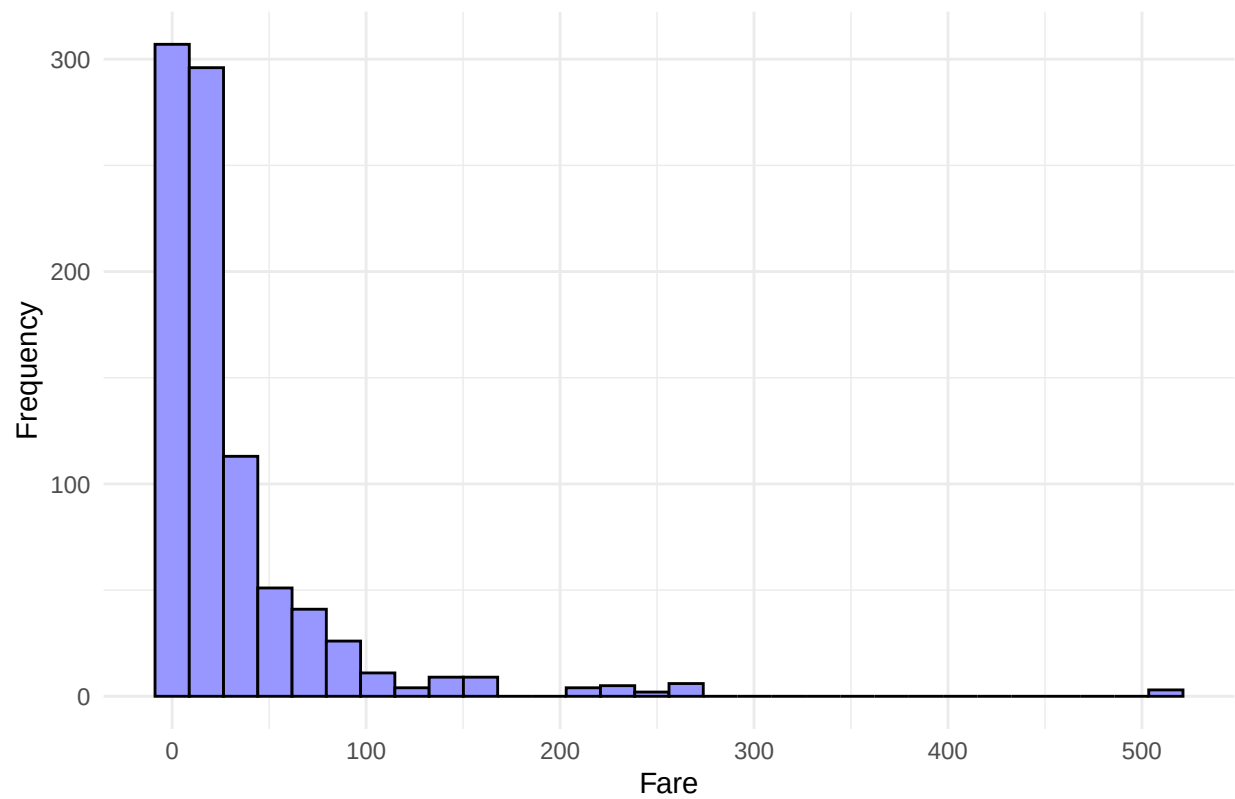
To gauge the economic distribution of Titanic passengers, we can look at the distribution of ticket fares (a) and passenger class (b):

```

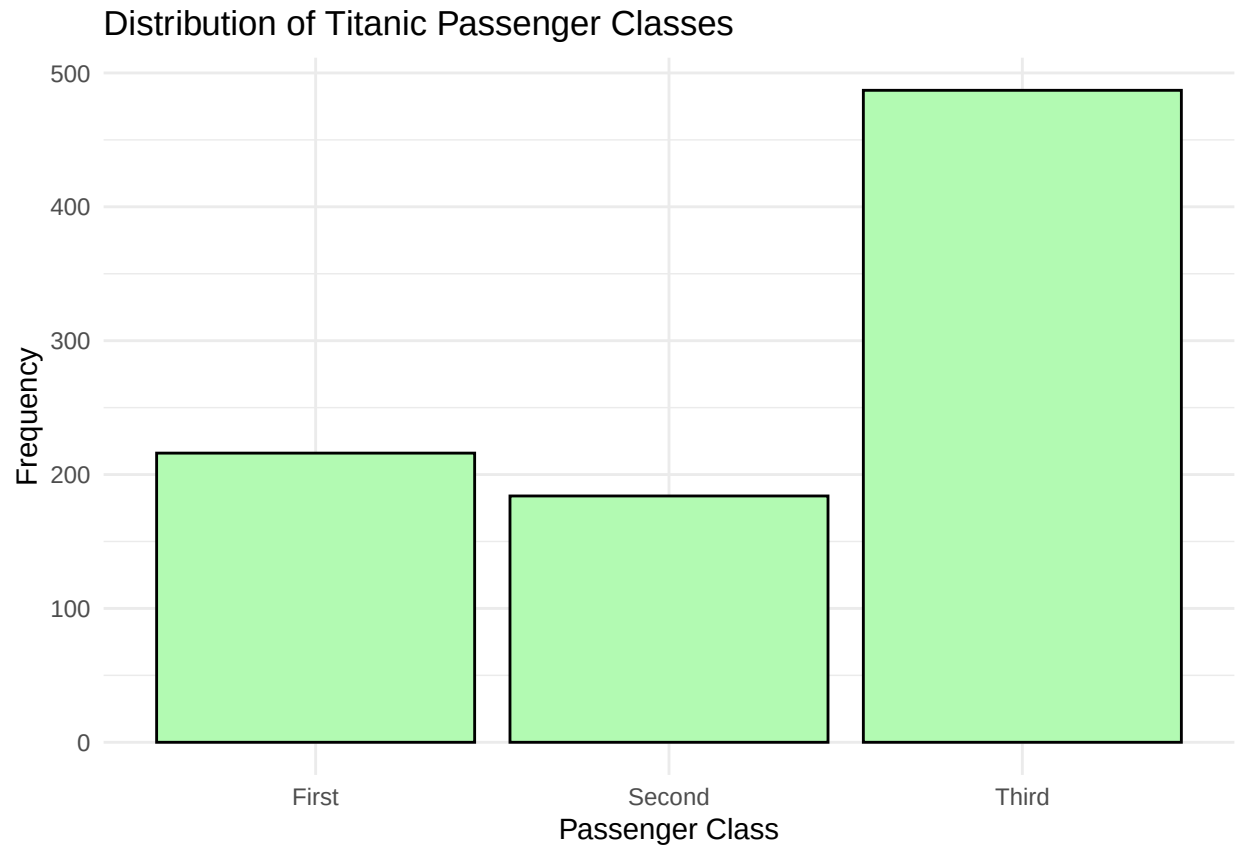
# Histogram of ticket fares
ggplot(titanic_clean, aes(x = Fare)) +
  geom_histogram(bins = 30, fill = "#9797ff", color = "black") +
  labs(title = "Distribution of Titanic Ticket Fares",
       x = "Fare",
       y = "Frequency") +
  theme_minimal()

```

Distribution of Titanic Ticket Fares



```
# Bar plot of passenger class
ggplot(titanic_clean, aes(x = Pclass)) +
  geom_bar(fill = "#B2FAB2", color = "black") +
  labs(title = "Distribution of Titanic Passenger Classes",
       x = "Passenger Class",
       y = "Frequency") +
  theme_minimal()
```

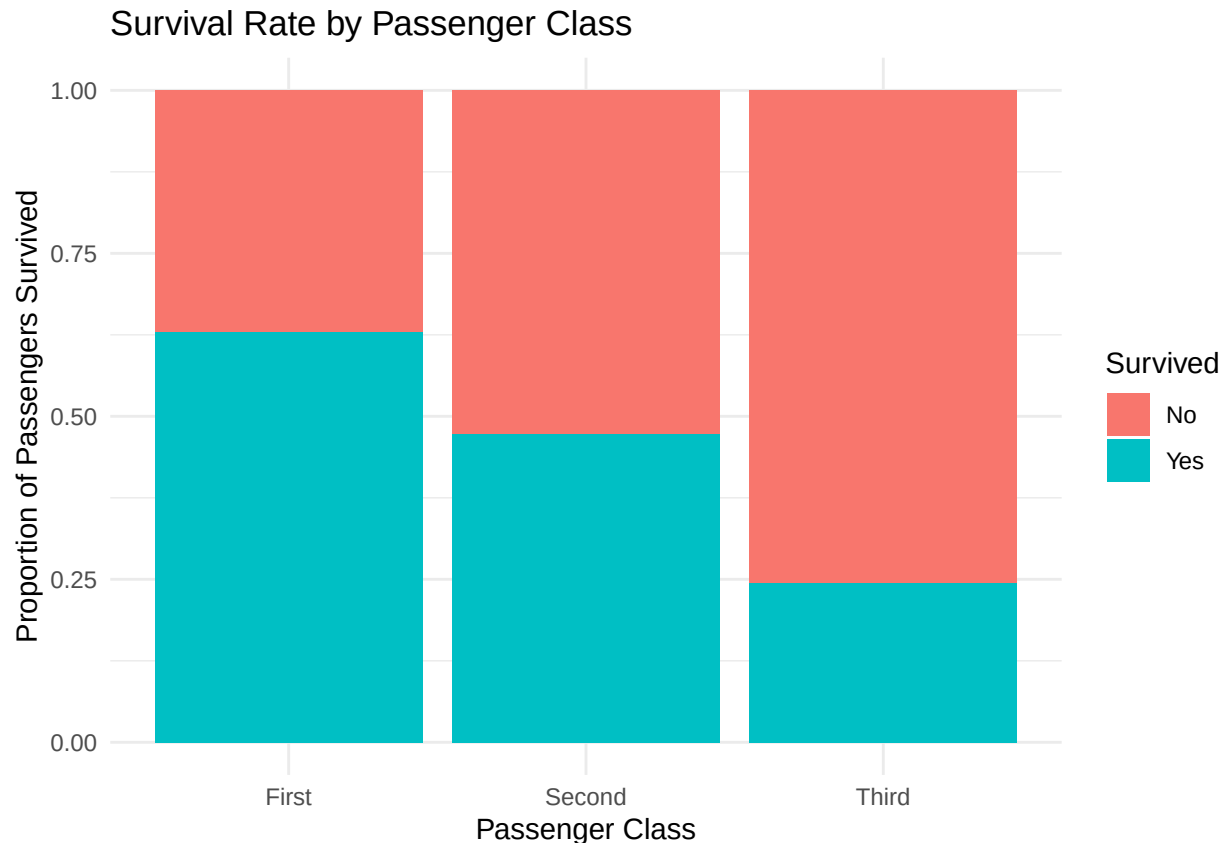


Analysis In 1912, the year of the Titanic, the American dollar was worth roughly 1/30 of what it is today. So rather than judging the exact dollar amounts of the fares, we will focus on the fare distribution, where the highest end is assumed to be the most glamorous, and the lowest end the most affordable. Interestingly, while a wide range of fares is evident, the majority of tickets observed in this dataset are in the lower end of the spectrum. Likewise, the passenger class bar plot shows that the vast majority of passengers were in the third class.

2. Does passenger class correlate with survival?

To assess this, we plot the survival rate (from 0.00 to 1.00) by passenger class:

```
# Bar plot of survival rate by passenger class
ggplot(titanic_clean, aes(x = Pclass, fill = Survived)) +
  geom_bar(position = "fill") +
  labs(title = "Survival Rate by Passenger Class", x = "Passenger Class", y = "Proportion of Passengers")
theme_minimal()
```



Analysis The survival-class correlation is clearly illustrated through the above bar plot, which shows each class's survival rate side by side.

In the plot, the survival rate clearly increases as the class number decreases (eg. as the passenger class gets fancier). Therefore, there is a clear correlation here: the fancier the class, the higher the survival rate.

However, it is important to remind readers that correlation is not causation. For example, an external factor, such as the location of each class with respect to the lifeboats, could theoretically have impacted survival rates. If so, that could also explain this correlation, yet it could have been entirely coincidental and unrelated to class. This is just a theoretical example to illustrate the difference between correlation and causation.

3. Does ticket fare correlate with survival?

To answer this question, we will group ticket fares into bins and plot the survival rate by each bin. Above each bar, we will also show the number of passengers in that fare bin:

```
# Define the number of bins
number_of_bins <- 6

# Calculate the range of fares and define equal-width bins
fare_range <- range(titanic_clean$Fare, na.rm = TRUE)
fare_bins <- seq(from = fare_range[1], to = fare_range[2], length.out = number_of_bins + 1)

# Add a new column for fare bins to the dataset
```

```
titanic_clean$FareBin <- cut(titanic_clean$Fare, breaks = fare_bins, include.lowest = TRUE, right = TRUE)

# Create a summary table of survival rate by fare bin, including count of passengers
fare_survival_summary <- titanic_clean %>%
  group_by(FareBin) %>%
  summarise(SurvivalRate = mean(as.numeric(Survived) - 1),
            Count = n(),
            .groups = 'drop')

# Plot survival rate by fare bin with count labels
ggplot(fare_survival_summary, aes(x = as.factor(FareBin), y = SurvivalRate)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  geom_text(aes(label = Count), vjust = -0.5, color = "black") +
  labs(title = "Survival Rate by Ticket Fare Bins",
       x = "Fare Bin (in USD)",
       y = "Survival Rate",
       caption = "Note: Numbers above bars represent total passengers in each fare bin.") +
  theme_minimal() +
  scale_y_continuous(labels = percent_format(scale = 100))
```



Analysis The data in this plot appears to be skewed - with a general trend of increasing survival rate as fare increases - although it is not 100% conclusive from this graph alone. For example, one may be shocked that the last bin, with the highest-paying passengers, had a 100% survival rate. However, it had a very small sample size (just 3 passengers)! While one may argue that this is highly statistically significant, since the

overall survival rate is only about 38%, it is still important to note the limitations of this particular analysis. We would need to study the methodology behind the data collection process of this dataset, and perhaps analyze (if theoretically available) a full dataset of all Titanic passengers to draw more conclusive results.

4. Does family size correlate with survival?

The dataset contains the following variables related to family size aboard:

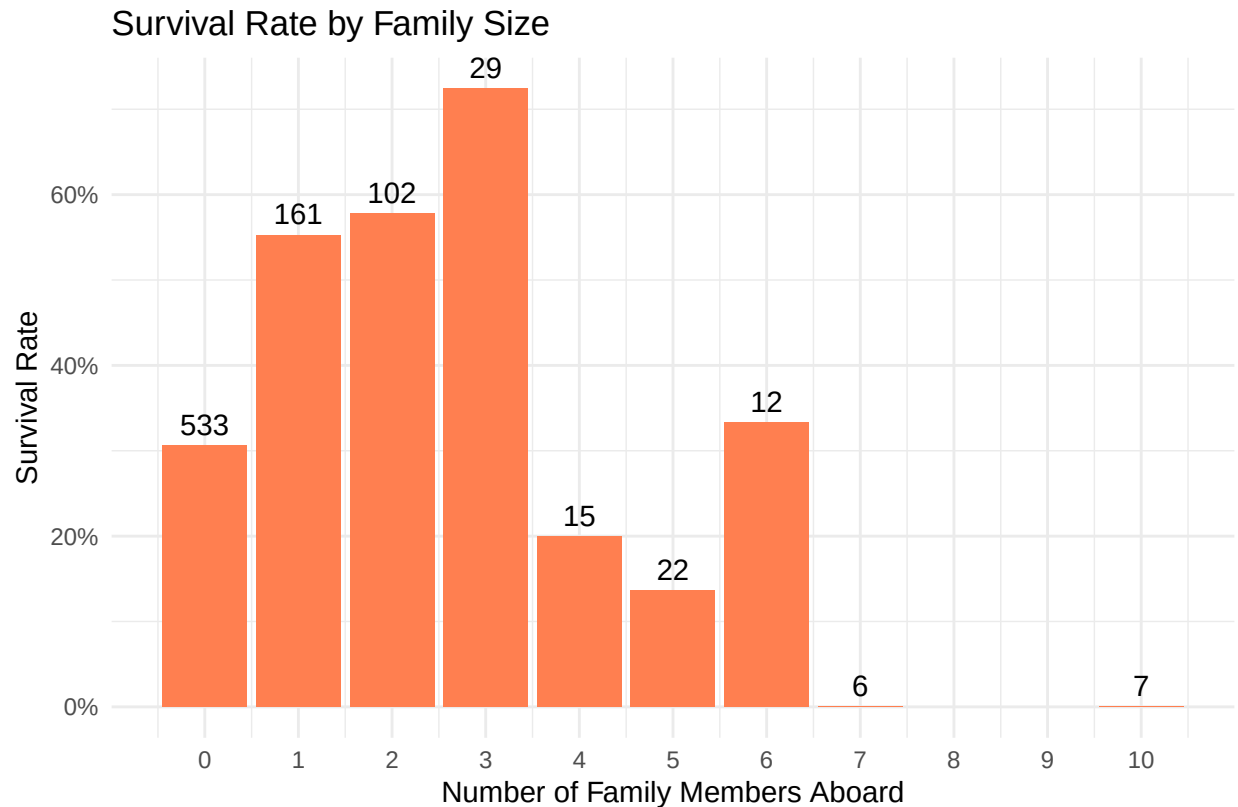
- Siblings.Spouses.Aboard
- Parents.Children.Aboard

We will calculate the passengers' family size aboard by summing together these variables. Then, we will calculate the survival rate against the family size and plot it in a bar graph. The total number of passengers in each family size group will be shown above the bars. This should give us a sense if family size aboard has a strong correlation with survival.

```
# Calculate total family size aboard for each passenger
titanic_clean <- titanic_clean %>%
  mutate(family_size = Siblings.Spouses.Aboard + Parents.Children.Aboard)

# Create a summary table of survival rate by family size
family_survival_summary <- titanic_clean %>%
  group_by(family_size) %>%
  summarise(
    SurvivalRate = mean(as.numeric(Survived) - 1),
    TotalPassengers = n(),
    .groups = 'drop'
  )

# Plot survival rate by family size with count labels
ggplot(family_survival_summary, aes(x = family_size, y = SurvivalRate)) +
  geom_bar(stat = "identity", fill = "coral") +
  geom_text(aes(label = TotalPassengers), vjust = -0.5, color = "black") +
  labs(title = "Survival Rate by Family Size",
       x = "Number of Family Members Aboard",
       y = "Survival Rate",
       caption = "Note: Numbers above bars represent total passengers in each family size group.") +
  theme_minimal() +
  scale_x_continuous(breaks = seq(0, 10, by = 1)) +
  scale_y_continuous(labels = percent_format(scale = 100))
```



Note: Numbers above bars represent total passengers in each family size group.

Analysis The plot shows that there is no linear correlation between family size aboard and survival rate. The highest survival rate observed is for the family size of 3, with over 70% surviving. The family sizes below that had decreased survival rates, and the lowest (0%) is observed for family sizes of 7 and 10. There might be a nonlinear correlation, but it is not confirmed from this graph alone, and would require further analysis.

5. Did a particular age group have the highest survival rate?

To answer this question, we will group the passengers into age groups and calculate the survival rate for each. We will then plot the survival rate against the age groups.

The following age groups will be used:

- 0-12
- 13-19
- 20-59
- 60+

```
# Define age groups
titanic_clean <- titanic_clean %>%
  mutate(age_group = case_when(
    Age <= 12 ~ "0-12",
    Age <= 19 ~ "13-19",
    Age <= 59 ~ "20-59",
    Age > 59 ~ "60+"
  ))
```

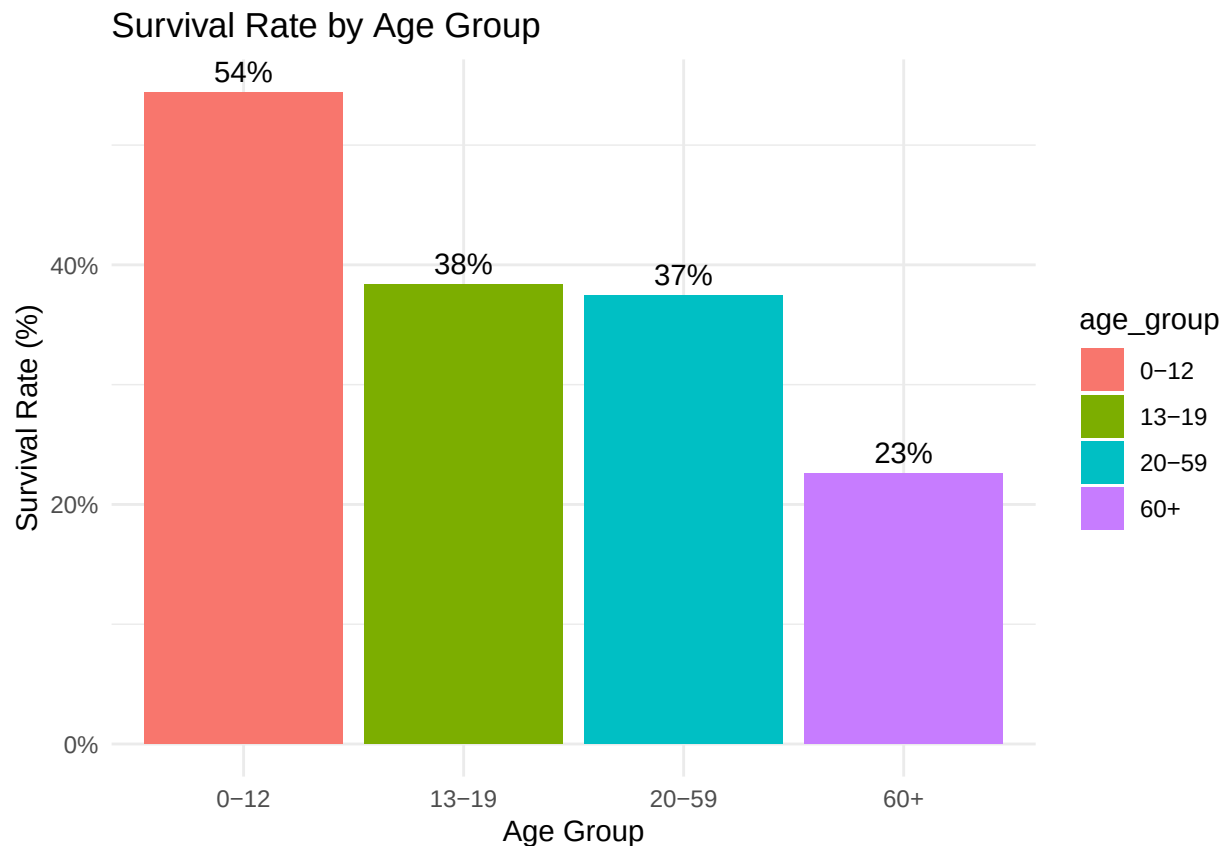
```

    TRUE ~ "60+"
  ))

# Calculate survival rate by age group
age_group_survival <- titanic_clean %>%
  group_by(age_group) %>%
  summarise(
    SurvivalRate = mean(as.numeric(Survived) - 1),
    Count = n(),
    .groups = 'drop'
  )

# Plot survival rate by age group
ggplot(age_group_survival, aes(x = age_group, y = SurvivalRate, fill = age_group)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = scales::percent(SurvivalRate, accuracy = 1)), vjust = -0.5) +
  labs(title = "Survival Rate by Age Group",
       x = "Age Group",
       y = "Survival Rate (%)") +
  theme_minimal() +
  scale_y_continuous(labels = scales::percent_format(scale = 100))

```



Analysis The plot clearly shows that the survival rate decreases as the age group increases. The highest survival rate is observed for the 0-12 age group, followed by the 13-19 age group, which is just 1% higher

than the 20-59 age group. The 60+ age group has a significant drop in survival rate, with only 23% of aged 60+ passengers surviving.

Implications

The following is a summary of our findings:

- There was a wide range of ticket fares and passenger classes aboard the Titanic, although the majority were on the lower end of the spectrum.
- There is a strong correlation between passenger class and survival rate: the fancier the class, the higher the survival rate.
- There is a correlation between ticket fare and survival rate: the higher the fare, the higher the survival rate.
- There is no linear correlation between family size and survival rate, although a more complex model may be able to identify a nonlinear correlation or pattern.
- There is a strong linear correlation with age and survival rate: the youngest age group had the highest survival rate, and it decreased as age increased.

These findings imply that socioeconomic status indeed may have played a role in the survival outcomes of the Titanic's passengers. However, we make that suggestion with the reservation that correlation does not imply causation, and other external factors may have also been involved.

Limitations

The scope of this study was limited by the following factors:

- Sample size: The dataset only contains 887 passengers, about 40% of the total 2,224 passengers who were on the Titanic.
- Unknown data collection methodology: The data source does not specify the exact methodology used for the collection of this dataset.
- We only investigated correlation amongst variables, and correlation does not imply causation.
- Only linear correlation was addressed, leaving out the possibility of more complex correlations and impact of external factors.
- We did not standardize and compare the variables' impacts on survival rate between each other.

Concluding Remarks

This study explored the Titanic dataset provided by Stanford, in attempt to shed light on the problem statement of whether or not socioeconomic factors played a role in survival outcomes.

Socioeconomic variables such as passenger class and ticket fare were indeed found to linearly correlate with survival rate, although correlation is not causation. Some fascinating insights were discovered, such as the strong correlation between age and survival rate, as well as potentially more complex patterns hinted to by the shapes of the plots above. I recommend a linear regression model to address the problem statement, and believe it would be effective at highlighting specific predictors in the data which contributed to the survival outcomes, and the importance of socioeconomic status with respect to other factors on survival.

Despite several noted limitations, this analysis brought out some interesting insights in Stanford's Titanic dataset, and laid the groundwork for earning more clarity on the problem statement via a logistic regression model.